MATH I (Fall 2026)

WEEK 5: INDEPENDENT STUDY ASSIGNMENTS (Homework)

About the Independent Study Assignments

The Independent Study Assignments are homework problems and assignments that will help you learn the week's content more deeply by solving problems and expanding your knowledge and skills. *Your answers will not be submitted and will not be graded*. You may solve the problems individually or collaborate with your fellow students or ask the assistance from an AI tool (to help you clarify the concepts, not providing the solution without you thinking about it first).

Week 5 - Assignments

The following assignments cover the material presented in Week 5.

Assignment 1. [*Optional assignment*]

The following is an *optional assignment – you can engage with it if you are familiar with AI-assistants, or if you want to start learning what AI-assistants can do for you in a context-specific domain*. The exercise is as follows:

Interact with an AI-assistant (such as ChatGPT, Claude, DeepSeek, Gemini, etc.), by structured conversation, through the following steps:

Think of the most important points of the lecture's content and ask the AI-assistant to evaluate your answer (you may ask the AI-assistant to evaluate it (usually, they do the evaluation in terms of content and clarity, but you can also ask for depth and complexity, or other criteria you may choose). The AI-assistant will not grade your input but it will provide content and perspectives to help you better understand the material.

If you find wrong outcomes, discrepancies or contradictory statements, converse with the AI-assistant to mention these points, asking for explanations, to resolve them. If you cannot resolve the wrong outcomes, please consult the instructor.

Assignment 2:

Study well **all** the problems, exercises, and examples that were presented in Lecture and Practical Session.

Assignment 3:

From Section 2.4 of the textbook (OpenStax Calculus 1), solve the following exercises:

131-136, 138, 145, 147, 149

Assignment 4:

Consider

$$s(t) = \sin\left(\frac{1}{t}\right), \qquad t \neq 0.$$

Can $s(0)$ be defined so that s is continuous at $t = 0$?

Assignment 5:

Consider

$$y(t) = \begin{cases} t\sin(1/t), & t \neq 0, \\ 0, & t = 0. \end{cases}$$

Is y continuous at $t = 0$?

Assignment 6: Designing a continuous sensor model

A sensor uses the model

$$S(x) = \begin{cases} ax + b, & x < 1, \\ 3, & x = 1, \\ x^2 + 2, & x > 1. \end{cases}$$

Find the condition on a and b that makes S continuous at $x = 1$. Give two possible pairs (a, b).

In [ ]:

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